

Continuous-Time Linear Time-Invariant (LTI) System

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October 4, 2026

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1 Continuous-Time Linear Time-Invariant (LTI) System

I Definition

i Causal continuous-time linear time-invariant (LTI) system

A causal continuous-time linear time-invariant (LTI) system of a variable $t \in \mathbb{R}$ is a system that accept an input (distribution) (often called function) $x(t)$ and produces a response (distribution) (often called function) $y(t)$ that is the solution of a linear autonomous ODE whose forcing term is $x(t)$, called governing equation, given initial condition $y(0) = 0$. It can be completely characterized by a distribution called impulse response (function) $h(t)$, which is the output when the input is the Dirac delta function $\delta(t)$ and is such that $h(t) = 0$ for all $t < 0$, that

$$y(t) = (x * h)(t) := \int_0^t x(\tau)h(t - \tau) d\tau.$$

Applying unilateral Laplace transform

$$Y(s) = X(s) \cdot H(s),$$

where $H(s)$ is a rational function called transfer function (轉移函數). Unless otherwise specified, we will discuss casual continuous-time linear time-invariant (LTI) system below.

ii Non-causal continuous-time linear time-invariant (LTI) system

A non-causal continuous-time linear time-invariant (LTI) system of a variable $t \in \mathbb{R}$ is a system that accept an input (distribution) (often called function) $x(t)$ and produces a response (distribution) (often called function) $y(t)$ that is the solution of a linear autonomous ODE whose forcing term is $x(t)$, called governing equation. It can be completely characterized by a distribution called impulse response (function) $h(t)$, which is the output when the input is the Dirac delta function $\delta(t)$, that

$$y(t) = (x * h)(t) := \int_{-\infty}^{\infty} x(\tau)h(t - \tau) d\tau.$$

Applying bilateral Laplace transform

$$Y(s) = X(s) \cdot H(s),$$

where $H(s)$ is a rational function called transfer function.

iii Gain

$|H(s)|$ is called gain (增益), or attenuation (衰減量) if < 1 .

iv Order

The number of poles of $H(s)$ is called the order of the system.

II Response

i Poles and zeros

Since $H(s)$ is rational, a complex conjugate of a pole is a pole of same order, and a complex conjugate of a zero is a zero of same order.

A pole-zero plot or pole-zero constellation is a graphical representation showing poles and zeros of a transfer function in the complex plane.

ii Transient response

A response component that converges to an equilibrium as t gets large. Such converged state is called steady state or transient state.

iii Time constant

For a response in the form

$$y(t) = A + \sum_{j=1}^n B_j e^{-\frac{t}{\tau_j}}$$

with $A \in \mathbb{R}$, $B_j \neq 0$, and $\tau_j > 0$.

Each τ_j is called a time constant, and the system is called a n -time-constant system, e.g., single-time-constant (STC), two-time-constant (TTC).

For each term

$$y_j(t) = B_j e^{-\frac{t}{\tau_j}}$$

if $B_j > 0$, we typically call that

- $y_j(t)$ decays to 0 about $5\tau_j$.
- The falling time or pulse decay time is defined as the time $y_j(t)$ changes from $0.9B_j$ to $0.1B_j$, which is $\ln 9\tau_j \approx 2.2\tau$.

if $B_j < 0$, we typically call that

- $y_j(t)$ grows to B_j about $5\tau_j$.
- The rising time or pulse grow time is defined as the time $y_j(t)$ changes from $0.1B_j$ to $0.9B_j$, which is $\ln 9\tau_j \approx 2.2\tau$.

iv Natural frequency

For a pair of conjugate poles $\pm i\omega$ of $H(s)$, ω is called a natural frequency of the system.

v Resonance

Resonance occurs when the input and transfer function have common poles on the imaginary axis. Such natural frequency is called resonant/resonance frequency.

III Stability

i Asymptotically stable or bounded-input bounded-output (BIBO) stable

A LTI system is called asymptotically stable or bounded-input bounded-output stable iff bounded input implies bounded response, iff all poles of it are on the left half-plane iff $h(t)$ is absolutely integrable.

ii (Lyapunov) stable

A LTI system is called (Lyapunov) stable iff there is no pole of it on the right half-plane and no multiple pole of it on the imaginary axis iff $h(t)$ is bounded.

iii Marginally stable

A LTI system is called marginally stable iff it is stable but not asymptotically stable.

iv Unstable

A LTI system is called unstable iff it is not stable.

IV Frequency response

i Frequency response

- $H(i\omega)$ is called frequency response,
- $a(\omega) := |H(i\omega)|$ is called magnitude/amplitude response/ratio or gain,
- $\theta(\omega) := \arg(H(i\omega))$ is called phase (angle) response/shift.

ω is called angular frequency and $f := \frac{\omega}{2\pi}$ is called non-angular frequency. Both can be called frequency and can be differentiated with symbol, and angular frequency is used below.

ii Steady-state sinusoidal response

Assume the system is stable with m LHP poles $a_1, \dots, a_n$ and n pure imaginary poles $b_1, \dots, b_n$ and that $H(s) = O(1)$. When the input is $x(t) = e^{i\omega t}$, $X(s) = \frac{1}{s - i\omega}$, the response is

$$Y(s) = \frac{H(s)}{s - i\omega}.$$

$$y(t) = \frac{1}{2\pi i} \lim_{T \rightarrow \infty} \int_{\gamma - iT}^{\gamma + iT} Y(s) e^{st} ds, \quad \gamma > 0.$$

Close the contour from $\gamma - iT$ to $\gamma + iT$ with a semicircular contour C_T

$$C_T = \{\gamma + iT e^{i\theta} : \theta \in [0, \pi]\}$$

to form a closed contour Γ_T .

$$\begin{aligned} \left| \int_{C_T} Y(s) e^{st} ds \right| &= \left| \int_{C_T} Y(s) e^{(\gamma - T \sin \theta)t} e^{i \Im(s)t} ds \right| \\ &= \int_0^\pi |Y(\gamma + iT e^{i\theta}) e^{(\gamma - T \sin \theta)t} (-T e^{i\theta})| d\theta \\ &\leq T e^{\gamma t} \int_0^\pi |Y(\gamma + iT e^{i\theta})| e^{-T \sin \theta t} d\theta \end{aligned}$$

$$Y(s) = O\left(\frac{1}{s}\right)$$

Suppose for some $M > 0$, for all sufficiently large $|s|$,

$$|Y(s)| \leq \frac{M}{|s|}$$

For all sufficiently large T ,

$$\begin{aligned} |\gamma + iT e^{i\theta}| &\geq T - \gamma \\ |Y(\gamma + iT e^{i\theta})| &\leq \frac{M}{T - \gamma} = O\left(\frac{1}{T}\right) \\ T e^{\gamma t} \int_0^\pi |Y(\gamma + iT e^{i\theta})| e^{-T \sin \theta t} d\theta &= O\left(\int_0^\pi e^{-T \sin \theta t} d\theta\right) \\ \int_0^\pi e^{-T \sin \theta t} d\theta &= 2 \int_0^{\frac{\pi}{2}} e^{-T \sin \theta t} d\theta \\ \sin \theta &\geq \frac{2\theta}{\pi} \\ e^{-T \sin \theta t} &\leq e^{-\frac{2T\theta t}{\pi}} \\ 2 \int_0^{\frac{\pi}{2}} e^{-T \sin \theta t} d\theta &\leq 2 \int_0^{\frac{\pi}{2}} e^{-\frac{2T\theta t}{\pi}} d\theta \\ &= 2 \frac{\pi}{2Tt} \left\{ e^{-\frac{2T\theta t}{\pi}} \right\} \Big|_0^{\frac{\pi}{2}} \\ &= \frac{\pi}{Tt} (1 - e^{-Tt}) \\ &\leq \frac{\pi}{Tt} \end{aligned}$$

$$\begin{aligned} 2 \int_0^{\frac{\pi}{2}} e^{-T \sin \theta t} d\theta &= O\left(\frac{1}{T}\right) \\ T e^{\gamma t} \int_0^\pi |Y(\gamma + iT e^{i\theta})| e^{-T \sin \theta t} d\theta &= O\left(\frac{1}{T}\right) \end{aligned}$$

Thus,

$$\begin{aligned} \lim_{T \rightarrow \infty} \int_{C_T} Y(s) e^{st} ds &= 0 \\ y(t) &= \frac{1}{2\pi i} \lim_{T \rightarrow \infty} \int_{\Gamma} Y(s) e^{st} ds \end{aligned}$$

Since $H(s)$ is meromorphic, by residue theorem,

$$y(t) = \text{Res}(Y(s) e^{st}, i\omega) + \sum_{k=1}^m \text{Res}(Y(s) e^{st}, a_k) + \sum_{k=1}^n \text{Res}(Y(s) e^{st}, b_k)$$

Use Laurent series:

$$\text{Res}(Y(s)e^{st}, z) = \lim_{s \rightarrow z} (s - z)Y(s)e^{st} = e^{zt} \lim_{s \rightarrow z} (s - z)Y(s)$$

$\sum_{k=1}^m \text{Res}(Y(s)e^{st}, a_k)$ is transient response.

- If $n = 0$, the system is asymptotically stable, and

$$\text{Res}(Y(s)e^{st}, i\omega) = H(i\omega)e^{i\omega t}$$

is steady-state sinusoidal response with amplitude $a(\omega)$.

- If $n > 0$, the system is marginally stable, and:

- If $i\omega$ is not a pole,

$$\text{Res}(Y(s)e^{st}, i\omega) + \sum_{k=1}^n \text{Res}(Y(s)e^{st}, b_k) = H(i\omega)e^{i\omega t} + \sum_{k=1}^n e^{b_k t} \lim_{s \rightarrow b_k} (s - b_k)Y(s)$$

is steady-state sinusoidal response and the second term is not transient.

- If $i\omega$ is a pole, it is resonant response that grows to infinity over time.

iii Logarithmic scale, decibel (dB), and Bode plot (波德圖)

Frequency response are often expressed in logarithmic scale:

- Magnitude response, input, and output can be expressed in decibel (dB) as $g(\omega) := C \log_{10}|H(i\omega)|$, $C \log_{10}|x|$, and $C \log_{10}|y|$ with some constant C . $C = 20$ is used here.
- Changing by 2 times is called an octave or oct, and changing by 10 times is called a decade or dec. $k \text{ dec}^{-1}$ equals to $\log_{10}(2) \text{ oct}^{-1}$. The most common used is 20 dB/dec equals to $20 \log_{10}(2) \approx 6 \text{ dB/oct}$.
- Bode magnitude plot or dB gain curve: Magnitude response plotted with frequency in logarithmic scale as horizontal axis and magnitude response in dB as vertical axis.
- Bode phase plot or phase curve: Phase response plotted with frequency in logarithmic scale as horizontal axis and phase response as vertical axis.
- Asymptotic Bode plot: The asymptotes of Bode plots for low and high frequency is often interested.

We can factor $H(s)$ into

$$H(s) = G \prod_{k=1}^m H_k(s)$$

then

$$a(\omega) = |G| \prod_{k=1}^m |H_k(i\omega)|$$

$$g(\omega) = 20 \log_{10} |G| + 20 \sum_{k=1}^m \log_{10} |H_k(i\omega)|$$

$$\theta(\omega) = \arg(G) + \sum_{k=1}^m \arg(H_k(i\omega))$$

iv Cutoff (截止), break, or corner (angular) frequency

For a transfer function, cutoff frequency ω_c is a frequency such that

$$|H(i\omega_c)| = \frac{\max_{\omega} a(\omega)}{\sqrt{2}}.$$

$20 \log_{10}\left(\frac{1}{\sqrt{2}}\right) = -10 \log_{10}(2) \approx -3$, thus this frequency is also denoted $\omega_{-3\text{dB}}$ or $\omega_{3\text{dB}}$.

v Normalizing

For a transfer function $H(s)$ such that

$$\max_{\omega} a(\omega) = K \in \mathbb{R}_{>0},$$

normalizing means dividing $H(s)$ by K . The following filter definitions are normalized. Non-normalized definition is defined as any filter whose normalized form complies with the definition.

vi Low-pass or lowpass (低通) (LP), high-cut, or bass-cut filter (LPF)

A continuous-time linear time-invariant system is a normalized low-pass filter iff $H(i\omega)$ as a function of a real variable ω is such that

$$\lim_{\omega \rightarrow 0} a(\omega) = 1$$

$$\lim_{\omega \rightarrow \infty} a(\omega) = 0$$

K is called low-pass gain.

Ideal normalized low-pass filter is

$$H(i\omega) = \begin{cases} 1, & \omega < \omega_0 \\ 0, & \omega > \omega_0 \end{cases}$$

vii High-pass or highpass (高通) (HP), low-cut, or treble-cut filter (HPF)

A continuous-time linear time-invariant system is a normalized high-pass filter iff $H(i\omega)$ as a function of a real variable ω is such that

$$\lim_{\omega \rightarrow 0} a(\omega) = 0$$

$$\lim_{\omega \rightarrow \infty} a(\omega) = 1$$

K is called high-pass gain.

Ideal normalized high-pass filter is

$$H(i\omega) = \begin{cases} 1, & \omega > \omega_0 \\ 0, & \omega < \omega_0 \end{cases}$$

viii Band-pass or bandpass (BP) filter (BPF)

A continuous-time linear time-invariant system is a standard normalized band-pass filter iff $H(i\omega)$ as a function of a real variable ω is such that

$$\begin{aligned}\lim_{\omega \rightarrow 0} a(\omega) &= 0 \\ \lim_{\omega \rightarrow \infty} a(\omega) &= 0 \\ \arg_{\omega} \left\{ a(\omega) = \frac{1}{\sqrt{2}} \right\} &= \omega_L, \omega_H \\ |H(\sqrt{\omega_L \omega_H})| &= 1\end{aligned}$$

where $\omega_L < \omega_H$ are called band edge, $\omega_H - \omega_L$ is called bandwidth, and $\omega_0 = \sqrt{\omega_L \omega_H}$ is called central passed frequency or center frequency. $\omega_0 \gg B$ is called narrow band-pass filter.

K is called band-pass or midband gain.

Ideal normalized band-pass filter is

$$H(i\omega) = \begin{cases} 1, & \omega_H > \omega > \omega_L \\ 0, & \omega < \omega_L \vee \omega > \omega_H \end{cases}$$

ix Band-stop or bandstop (BS), band-reject, or notch filter (BSF)

A continuous-time linear time-invariant system is a standard normalized band-stop filter iff $H(i\omega)$ as a function of a real variable ω is such that

$$\begin{aligned}\arg_{\omega} \left\{ a(\omega) = \frac{1}{\sqrt{2}} \right\} &= \omega_L, \omega_H \\ |H(\sqrt{\omega_L \omega_H})| &= 0\end{aligned}$$

where $\omega_L < \omega_H$ are called band edge, $\omega_H - \omega_L$ is called bandwidth, and $\omega_0 = \sqrt{\omega_L \omega_H}$ is called central rejected frequency or center frequency. $\omega_0 \gg B$ is called narrow band-stop filter.

K is called notch gain.

Ideal normalized band-stop filter is

$$H(i\omega) = \begin{cases} 0, & \omega_H > \omega > \omega_L \\ 1, & \omega < \omega_L \vee \omega > \omega_H \end{cases}$$

x Bandwidth

The band of frequencies over which the gain is no less than 3 dB (or other value if specified, such as 0 dB for unity gain bandwidth), or the length of it if it is connected.

xi Normalized low-pass prototype filter

A normalized low-pass prototype filter is a low-pass filter with cutoff frequency $\omega_c = 1$. Suppose the transfer function of it is $H_P(s)$. A normalized low-pass prototype filter is usually defined with $(a(\omega))^2 = H(i\omega)H(-i\omega)$ or $H(s)$.

To transform an n th-order normalized low-pass prototype filter to an n th-order low-pass filter with cutoff frequency ω_c , use the transfer function

$$H_{LP}(s) = H_P\left(\frac{s}{\omega_c}\right).$$

To transform an n th-order normalized low-pass prototype filter to an n th-order high-pass filter with cutoff frequency ω_c , use the transfer function

$$H_{HP}(s) = H_P\left(\frac{\omega_c}{s}\right).$$

To transform an n th-order normalized low-pass prototype filter to a $2n$ th-order standard band-pass filter with cutoff frequency ω_c , use the transfer function

$$H_{BP}(s) = H_P\left(\frac{s^2 + \omega_0^2}{Bs}\right).$$

To transform an n th-order normalized low-pass prototype filter to a $2n$ th-order standard band-stop filter with cutoff frequency ω_c , use the transfer function

$$H_{BS}(s) = H_P\left(\frac{Bs}{s^2 + \omega_0^2}\right).$$

xii Butterworth filter or maximally flat magnitude filter

n th-order Butterworth low-pass prototype filter is

$$(a(\omega))^2 = \frac{1}{1 + \omega^{2n}}$$

$$H(s) = \prod_{k=1}^n \frac{1}{s - e^{\frac{j(2k+n-1)\pi}{2n}}}$$

When not specified, a filter of a type usually refers to the lowest-order Butterworth filter of the type.

V Transfer function factors

Variables after ; are parameters, which are positive real numbers unless otherwise specified. Only positive ω are considered.

i Reciprocal factor

This factor is not asymptotically stable but marginally stable.

$$H_c(s; W) := \frac{W}{s}$$

$$H_c(i\omega; W) = \frac{W}{i\omega}$$

$$a_c(\omega; W) = \frac{W}{\omega}$$

$$g_c(\omega; W) = -20 \log_{10}(\omega) + 20 \log_{10}(W)$$

$$\theta_c(\omega; W) = -\frac{\pi}{2}$$

Bode magnitude plot has slope -20 dB/dec, y intercept $20 \log_{10}(W)$, and x intercept $-\log_{10}(W)$.

Limit:

- For $\omega \gg W$:

$$a_c(\omega) \approx 0$$

$$g_c(\omega) \approx -\infty$$

- For $\omega \ll W$:

$$a_c(\omega) \approx \infty$$

$$g_c(\omega) \approx \infty$$

ii Ramp factor

$$H_r(s; W) := \frac{s}{W}$$

$$H_r(i\omega; W) = \frac{i\omega}{W}$$

$$a_r(\omega; W) = \frac{\omega}{W}$$

$$g_r(\omega; W) = 20 \log_{10}(\omega) - 20 \log_{10}(W)$$

$$\theta_r(\omega; W) = \frac{\pi}{2}$$

Bode magnitude plot has slope 20 dB/dec, y intercept $-20 \log_{10}(W)$, and x intercept $\log_{10}(W)$.

Limit and cutoff frequency:

- For $\omega \gg W$:

$$a_r(\omega; W) \approx \infty$$

$$g_r(\omega; W) \approx \infty$$

- For cutoff frequency $\omega = \omega_c = \frac{W}{\sqrt{2}}$:

$$a_r(\omega; W) = \frac{1}{\sqrt{2}}$$

$$g_r(\omega; W) = -10 \log_{10}(2)$$

- For $\omega \ll W$:

$$a_r(\omega; W) \approx 0$$

$$g_r(\omega; W) \approx -\infty$$

iii Low-pass factor

$$H_{LP}(s; \omega_c) := \frac{\omega_c}{s + \omega_c}$$

$$H_{LP}(i\omega; \omega_c) = \frac{\omega_c}{i\omega + \omega_c}$$

$$a_{LP}(\omega; \omega_c) = \frac{\omega_c}{\sqrt{\omega^2 + \omega_c^2}}$$

$$g_{LP}(\omega; \omega_c) = 20 \log_{10}(\omega_c) - 10 \log_{10}(\omega^2 + \omega_c^2)$$

$$\theta_{LP}(\omega; \omega_c) = -\arg(i\omega + \omega_c)$$

Asymptotic Bode plot and cutoff frequency:

- For $\omega \gg \omega_c$:

$$a_{LP}(\omega; \omega_c) \approx \frac{\omega_c}{\omega}$$

$$g_{LP}(\omega; \omega_c) \approx 20 \log_{10}(\omega_c) - 20 \log_{10}(\omega),$$

which is slant asymptote with slope -20 dB/dec, y intercept $20 \log_{10}(\omega_c)$, and x intercept $\log_{10}(\omega_c)$.

$$\theta_{LP}(\omega; \omega_c) \approx 0$$

- For cutoff frequency $\omega = \omega_c$:

$$a_{LP}(\omega; \omega_c) = \frac{1}{\sqrt{2}}$$

$$g_{LP}(\omega; \omega_c) = -10 \log_{10}(2)$$

$$\theta_{LP}(\omega; \omega_c) = -\frac{\pi}{4}$$

- For $\omega \ll \omega_c$:

$$a_{LP}(\omega; \omega_c) \approx 1$$

$$g_{LP}(\omega; \omega_c) \approx 0,$$

which is horizontal asymptote $g_{LP} = 0$.

$$\theta_{LP}(\omega; \omega_c) \approx -\frac{\pi}{2}$$

iv High-pass factor

$$H_{HP}(s; \omega_c) := \frac{s}{s + \omega_c}$$

$$H_{HP}(i\omega; \omega_c) = \frac{i\omega}{i\omega + \omega_c}$$

$$a_{HP}(\omega; \omega_c) = \frac{\omega}{\sqrt{\omega^2 + \omega_c^2}}$$

$$g_{HP}(\omega; \omega_c) = 20 \log_{10}(\omega) - 10 \log_{10}(\omega^2 + \omega_c^2)$$

$$\theta_{HP}(\omega; \omega_c) = \frac{\pi}{2} - \arg(i\omega + \omega_c)$$

Asymptotic Bode plot and cutoff frequency:

- For $\omega \gg \omega_c$:

$$a_{LP}(\omega; \omega_c) \approx 1$$

$$g_{HP}(\omega; \omega_c) \approx 0,$$

which is horizontal asymptote $g_{HP} = 0$.

$$\theta_{HP}(\omega; \omega_c) \approx \frac{\pi}{2} - \frac{\pi}{2} = 0$$

- For cutoff frequency $\omega = \omega_c$:

$$a_{HP}(\omega; \omega_c) = \frac{1}{\sqrt{2}}$$

$$g_{HP}(\omega; \omega_c) = -10 \log_{10}(2)$$

$$\theta_{HP}(\omega; \omega_c) = \frac{\pi}{4}$$

- For $\omega \ll \omega_c$:

$$a_{HP}(\omega; \omega_c) \approx \frac{\omega}{\omega_c}$$

$$g_{HP}(\omega; \omega_c) \approx 20 \log_{10}(\omega) - 20 \log_{10}(\omega_c),$$

which is slant asymptote with slope 20 dB/dec, y intercept $-20 \log_{10}(\omega_c)$, and x intercept $\log_{10}(\omega_c)$.

$$\theta_{HP}(\omega; \omega_c) \approx \frac{\pi}{2}$$

v Quadratic low-pass factor

$$H_{QD}(s; \omega_0, Q) := \frac{\omega_0^2}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2}, \quad Q \geq \frac{1}{2}$$

$$H_{QD}(i\omega; \omega_0, Q) = \frac{\omega_0^2}{-\omega^2 + i\frac{\omega_0}{Q}\omega + \omega_0^2}$$

$$a_{QD}(\omega; \omega_0, Q) = \frac{\omega_0^2}{\sqrt{\frac{\omega_0^2}{Q^2}\omega^2 + \omega^4 + \omega_0^4 - 2\omega^2\omega_0^2}}$$

$$\frac{d}{d\omega} (a_{QD}(\omega; \omega_0, Q)) = \frac{-\frac{\omega_0^4}{Q^2}\omega - 2\omega^3\omega_0^2 + 2\omega\omega_0^4}{\left(\frac{\omega_0^2}{Q^2}\omega^2 + \omega^4 + \omega_0^4 - 2\omega^2\omega_0^2\right)^{\frac{3}{2}}}$$

$$\left. \frac{d}{d\omega} (a_{QD}(\omega; \omega_0, Q)) \right|_{\omega=\omega_p} = 0$$

$$\omega_p = \omega_0 \sqrt{1 - \frac{1}{2Q^2}}, \quad Q > \frac{1}{\sqrt{2}}$$

$$\theta_{QD}(\omega; \omega_0, Q) = -\arg\left(-\omega^2 + i\frac{\omega_0}{Q}\omega + \omega_0^2\right)$$

- If $Q \geq \frac{1}{2}$, it is a low-pass filter.

- If $Q > \frac{1}{2}$, the maximum of g_{QD} occurs at peak frequency $\omega = \omega_p = \omega_0 \sqrt{1 - \frac{1}{2Q^2}}$.
- If $Q = \frac{1}{2}$, at $\omega = \omega_0 \sqrt{1 - \frac{1}{2Q^2}} = 0$,

$$a_{QD}(\omega; \omega_0, Q) = \infty$$

$$g_{QD}(\omega; \omega_0, Q) = \infty$$

- If $Q < \frac{1}{2}$, it is unstable.

Asymptotic Bode plot and center and peak frequencies:

- For $\omega \ll \omega_0, \frac{\omega_0}{Q}$:

$$a_{QD}(\omega; \omega_0, Q) \approx 1$$

$$g_{QD}(\omega; \omega_0, Q) \approx 0,$$

which is horizontal asymptote $g_{QD} = 0$.

$$\theta_{QD}(\omega; \omega_0, Q) \approx 0$$

- For center frequency $\omega = \omega_0$:

$$|H_{QD}(i\omega; \omega_0, Q)| = Q$$

$$g_{QD}(\omega; \omega_0, Q) = 20 \log_{10}(Q)$$

$$\theta_{QD}(\omega; \omega_0, Q) = -\frac{\pi}{2}$$

- If $Q > \frac{1}{2}$, for peak frequency $\omega = \omega_p = \omega_0 \sqrt{1 - \frac{1}{2Q^2}}$:

$$a_{QD}(\omega; \omega_0, Q) = \frac{\omega_0^2}{\sqrt{\frac{\omega_0^4}{Q^2} - \frac{\omega_0^4}{2Q^4} + \omega_0^4 - \frac{\omega_0^4}{Q^2} + \frac{\omega_0^4}{4Q^4} + \omega_0^4 - 2\omega_0^4 + \frac{\omega_0^4}{Q^2}}} = \frac{Q}{\sqrt{1 - \frac{1}{4Q^2}}}$$

$$g_{QD}(\omega; \omega_0, Q) = 20 \log_{10}(Q) - 10 \log_{10}\left(1 - \frac{1}{4Q^2}\right),$$

which are maxima $a_{QD} = \frac{Q}{\sqrt{1 - \frac{1}{4Q^2}}}$ and $g_{QD} = 20 \log_{10}(Q) - 10 \log_{10}\left(1 - \frac{1}{4Q^2}\right)$.

- For $\omega \gg \omega_0, \frac{\omega_0}{Q}$:

$$a_{QD}(\omega; \omega_0, Q) \approx \frac{\omega_0^2}{\omega^2}$$

$$g_{QD}(\omega; \omega_0, Q) \approx 40 \log_{10}(\omega_0) - 40 \log_{10}(\omega)$$

which is slant asymptote with slope -40 dB/dec, y intercept $40 \log_{10}(\omega_0)$, and x intercept $\log_{10}(\omega_0)$.

$$\theta_{QD}(\omega; \omega_0, Q) \approx -\pi$$

vi Band-pass factor

$$H_{BP}(s; B, \omega_0) := \frac{Bs}{s^2 + Bs + \omega_0^2}$$

$$H_{BP}(i\omega; B, \omega_0) = \frac{iB\omega}{-\omega^2 + iB\omega + \omega_0^2}$$

$$a_{BP}(\omega; B, \omega_0) = \frac{B\omega}{\sqrt{B^2\omega^2 + \omega^4 + \omega_0^4 - 2\omega^2\omega_0^2}}$$

$$\frac{d}{d\omega} (a_{BP}(\omega; B, \omega_0)) = \frac{B(\omega^4 + \omega_0^4 - 2\omega^2\omega_0^2 + 2\omega^3 - 2\omega\omega_0^2)}{(B^2\omega^2 + \omega^4 + \omega_0^4 - 2\omega^2\omega_0^2)^{\frac{3}{2}}}$$

$$\left. \frac{d}{d\omega} (a_{BP}(\omega; B, \omega_0)) \right|_{\omega=\omega_0} = 0$$

$$\theta_{BP}(\omega; B, \omega_0) = \frac{\pi}{2} - \arg(-\omega^2 + iB\omega + \omega_0^2)$$

Asymptotic Bode plot and center and peak frequency:

- For $\omega \gg \omega_0, B$:

$$a_{BP}(\omega; B, \omega_0) \approx \frac{B}{\omega}$$

$$g_{BP}(\omega; B, \omega_0) \approx -20 \log_{10}(\omega) + 20 \log_{10}(B)$$

which is slant asymptote with slope -20 dB/dec, y intercept $20 \log_{10}(B)$, and x intercept $\log_{10}(B)$.

$$\theta_{BP}(\omega; B, \omega_0) \approx -\frac{\pi}{2}$$

- For center and peak frequency $\omega = \omega_0$:

$$a_{BP}(\omega; B, \omega_0) = 1$$

$$g_{BP}(\omega; B, \omega_0) = 0$$

which are maxima $a_{BP} = 1$ and $g_{BP} = 0$.

$$\theta_{BP}(\omega; B, \omega_0) = 0$$

- For $\omega \ll \omega_0, B$:

$$a_{BP}(\omega; B, \omega_0) \approx \frac{B\omega}{\omega_0^2}$$

$$g_{BP}(\omega; B, \omega_0) \approx 20 \log_{10}(\omega) + 20 \log_{10}(B) - 40 \log_{10}(\omega_0),$$

which is slant asymptote with slope 20 dB/dec, y intercept $20 \log_{10}(B) - 40 \log_{10}(\omega_0)$, and x intercept $2 \log_{10}(\omega_0) - \log_{10}(B)$.

$$\theta_{BP}(\omega; B, \omega_0) \approx \frac{\pi}{2}$$

vii Band-stop factor

$$H_{BS}(s; B, \omega_0) := \frac{s^2 + \omega_0^2}{s^2 + Bs + \omega_0^2}$$

$$H_{BS}(i\omega; B, \omega_0) = \frac{-\omega^2 + \omega_0^2}{-\omega^2 + iB\omega + \omega_0^2}$$

$$a_{BS}(\omega; B, \omega_0) = \frac{-\omega^2 + \omega_0^2}{\sqrt{B^2\omega^2 + \omega^4 + \omega_0^4 - 2\omega^2\omega_0^2}}$$

$$\theta_{BS}(\omega; B, \omega_0) = \arg\left(\frac{-\omega^2 + \omega_0^2}{-\omega^2 + iB\omega + \omega_0^2}\right)$$

Asymptotic Bode plot and center frequency:

- For $\omega \gg \omega_0, B$:

$$a_{BS}(\omega; B, \omega_0) \approx 1$$

$$g_{BS}(\omega; B, \omega_0) \approx 0,$$

which is horizontal asymptote $g_{BS} = 0$.

$$\theta_{BS}(\omega; B, \omega_0) \approx 0$$

- For center frequency $\omega = \omega_0$:

$$a_{BS}(\omega; B, \omega_0) = 0$$

$$g_{BS}(\omega; B, \omega_0) = -\infty$$

$$\theta_{BS}(\omega; B, \omega_0) = \arg(0) = \text{undefined}$$

- For $\omega \ll \omega_0, B$:

$$a_{BS}(\omega; B, \omega_0) \approx 1$$

$$g_{BS}(\omega; B, \omega_0) \approx 0$$

which is horizontal asymptote $g_{BS} = 0$.

$$\theta_{BS}(\omega; B, \omega_0) \approx 0$$